

FIFA 21 Player Dataset Analysis Report

Project Team

Student Name	Student ID
Mustafa Hussain Mohamed Hussain	23101483
Ahmed Elsadny Tawfik Elsadny	23101349
Roaya Ali el sayed Ali Badran	23101391

1. Executive Summary

This analysis aimed to reduce the dimensionality of the FIFA 21 player dataset to visualize player clusters and understand the underlying variance in player attributes. By processing raw data and applying Principal Component Analysis (PCA), we successfully retained **60.25% of the dataset's information in just 2 dimensions** and **72.38% in 3 dimensions**, allowing for effective visualization of player similarities.

2. Data Loading & Initial Inspection

- **Raw Data Source:** `fifa21_raw_data_v2.csv`
- **Initial Dimensions:** The dataset contained **18,979 rows** (players) and **77 columns** (attributes).
- **Initial Observation:** The raw data contained a mix of integers, strings (with symbols like '€', '★', 'lbs'), and missing values, requiring significant cleaning before analysis.

```
Shape: (18979, 77)
```

3. Data Cleaning & Preprocessing

To prepare the data for numerical analysis, the following transformation steps were executed:

- **Column Standardization:** All column names were converted to lowercase snake_case (e.g., `Release Clause` → `release_clause`) to ensure consistency.
- **Unit Conversions:**
 - **Height:** Converted from imperial (feet/inches) to metric (centimeters).
 - **Weight:** Converted from pounds (lbs) to kilograms (kg).
 - **Currency (value, wage, release_clause):** Stripped '€' symbols and converted 'M' (millions) and 'K' (thousands) suffixes into actual numerical float values.
 - **Social Metrics (hits):** Converted 'K' suffixes into thousands.
- **Symbol Removal:**

- Columns regarding player skills (`w/f`, `sm`, `ir`) contained '★' characters. These were removed, and the columns were converted to integers.
- Newlines were stripped from the `club` column.

4. Missing Value Imputation

- **Thresholding:** Columns with excessive missing data (more than 25% missing) were dropped to preserve data quality.
- **Imputation Strategy:**
 - **Numerical Data:** Remaining missing numerical values were filled using the **median** (to avoid skewing by outliers).
 - **Categorical Data:** Missing categorical values were filled using the **mode** (most frequent value).
- **Result:** The dataset was cleared of all null values (Remaining Missing Values: 0).

```
... Remaining Missing Values: 0
```

5. Feature Selection & Scaling

- **Feature Selection:** Non-predictive metadata columns were removed to focus solely on performance metrics.
 - *Dropped:* `id`, `name`, `longname`, `photourl`, `nationality`, `club`, `joined`, `contract`, `jersey_number`, etc.
- **Dimensionality:** After cleaning and dropping non-numerical columns, the dataset shape was **(18,979 rows, 60 columns)**.
- **Scaling:** The data was normalized using `StandardScaler`. This is a crucial step for PCA to ensure that variables with larger scales (like `value` in millions) do not dominate variables with smaller scales (like `age` or `skill_moves`).

```
... Data Scaled. Shape ready for PCA: (18979, 60)
```

6. Principal Component Analysis (PCA) Findings

PCA was applied to compress the 60 feature columns into fewer principal components (PCs) while retaining the maximum amount of information (variance).

A. 2-Dimensional Analysis

- **PC1 (Principal Component 1):** Explained **45.70%** of the variance.
 - *Interpretation:* This component likely captures the primary differentiator in football players, which is often the overall skill level or the separation between Goalkeepers and Field players.

- **PC2 (Principal Component 2):** Explained **14.55%** of the variance.
 - *Interpretation:* This likely differentiates distinct playstyles (e.g., defensive stats vs. attacking stats).
- **Total Information Retained (2D): 60.25%**
 - *Conclusion:* We can visualize the majority of the dataset's complexity using a simple X-Y scatter plot.

```
... Explained Variance (PC1): 45.70%  
    Explained Variance (PC2): 14.55%  
    Total Information Retained: 60.25%
```

B. 3-Dimensional Analysis

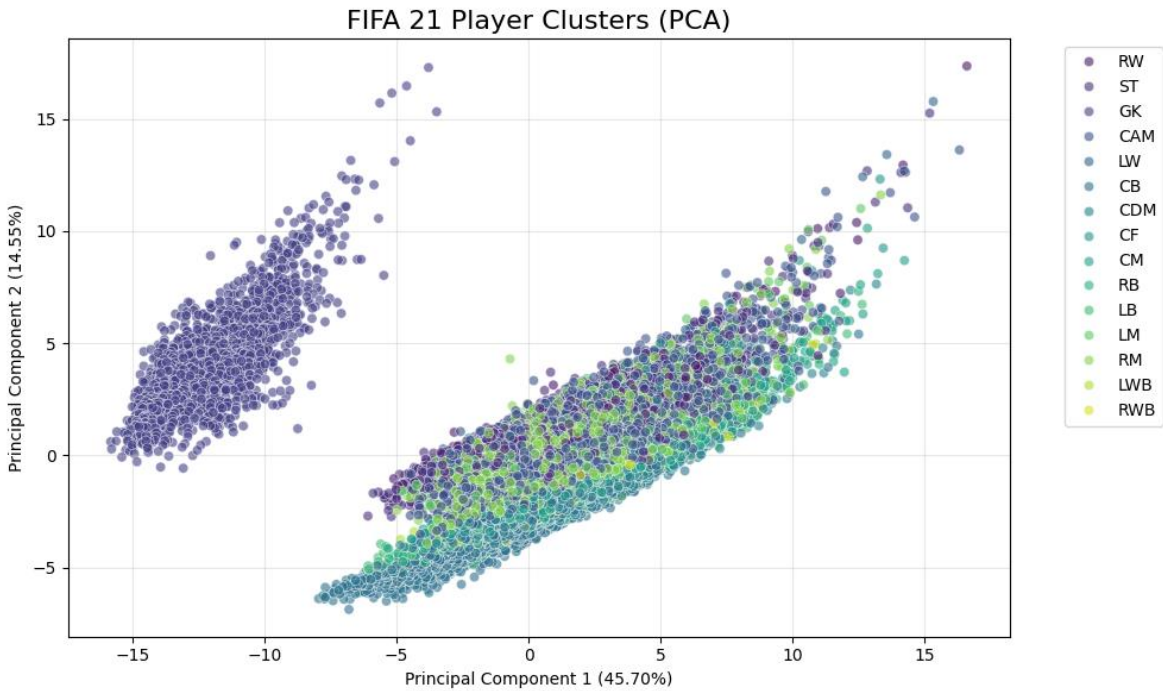
- **PC3 (Principal Component 3):** Added additional variance explanation.
- **Total Information Retained (3D): 72.38%**
 - *Conclusion:* Adding a third dimension significantly improves the model's ability to represent the players, capturing nearly three-quarters of the statistical difference between players.

```
... Total Variance in 3D: 72.38%
```

7. Visualizations Generated

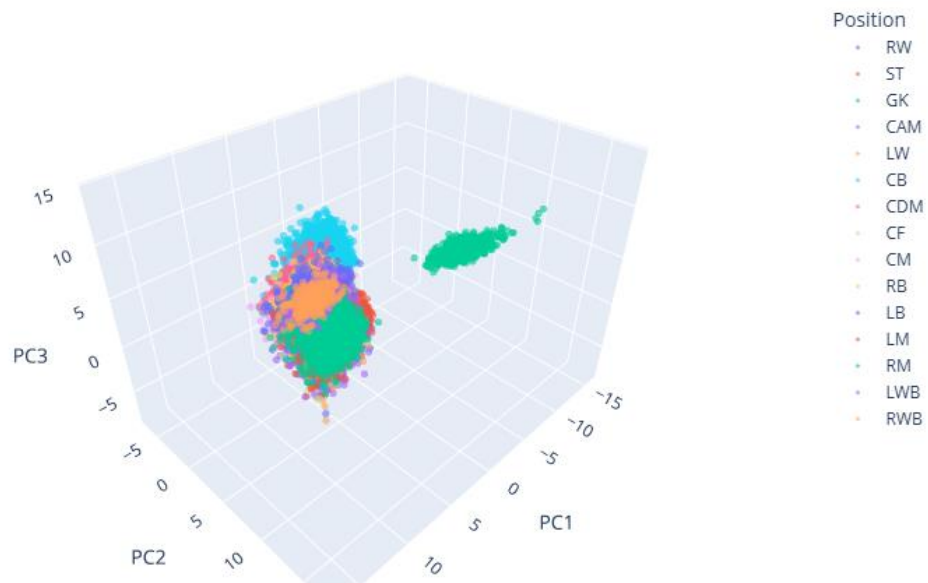
The code generated two visualizations (though not viewable in text output, their configuration implies the following):

1. **2D Scatter Plot:** A plot of PC1 vs. PC2, colored by `best_position`. This helps identify how different positions (Defenders, Midfielders, Forwards) cluster together naturally based on their stats.



2. **3D Interactive Plot:** A Plotly 3D scatter plot allowing for the exploration of PC1, PC2, and PC3, providing a deeper view of how player attributes segregate in 3D space.

FIFA 21 PCA Analysis (3D Interactive)



Conclusion

The preprocessing pipeline successfully converted raw, messy FIFA 21 data into a clean numerical format. The PCA analysis revealed that player attributes are highly correlated, as we were able to reduce 60 different metrics down to just 3 dimensions while still retaining **72.38%** of the information, providing a powerful way to cluster and analyze player types.